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# Navigating well-being in the digital era: A scoping review of digital well-being

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ABSTRACT

In today's hyperconnected world, most people across generations perceive frequent internet use as normal and acceptable. However, this assumption needs closer examination, as it overlooks potential risks associated with excessive connectivity—risks that may threaten our generation's well-being, known as *digital well-being* (DWB). The definition and formation of this construct remain fragmented. This study conducted a scoping review to identify, analyze, and synthesize existing definitions, measurements tools, and intervention strategies related to DWB. Using the Joanna Briggs Institute (JBI) methodology, seven major databases were systematically searched, yielding 26 eligible articles. Notably, research originating from Asia remains scarce. Existing definitions converge on three perspective: (1) DWB as an emerging concept of well-being influenced by technology use, (2) as the perceived impact of digital engagement on individual well-being, and (3) as a balanced-use strategy within digital environments. Three overarching themes were synthesized—behavioral intention to use technology, eudaimonic well-being, and socio-cultural contextual factors. Measurement tools of DWB remain limited; with five scales identified but none offering an integrated framework. Interventions, particularly those emphasizing digital disconnection, reveal mixed results regarding their effectiveness. Collectively, these findings highlight conceptual ambiguities, limited measurement development, and inconsistent intervention outcome. This review advances theoretical clarity by framing DWB as a multidimensional construct that bridges psychological and socio-technological domains. It also provide practical guidance for developing culturally adaptive interventions and policy frameworks that promote balanced and meaningful digital engagement.

## 1. Introduction

According to Data Reportal (2024), the number of internet users increased by 3.2 % from the previous year, reaching 5.56 billion users in 2024. This figure represents 67.9 % of the global population, although reporting delays suggest that the actual penetration rate may be even higher. The average duration of daily internet use has also grown steadily, reaching 6 h and 38 min. Internet engagement spans all age groups, though usage tends to decline with age.

The rise of digital connectivity has prompted growing concern about its potential negative consequences (Burr et al., 2018; Sharma et al., 2017). Research indicates that excessive smartphones (Ames, 2013) and social media use (Marino et al., 2018, 2020) can harm multiple aspects of daily life, including driving (Feldman et al., 2011), studying (Ames, 2013), and working (Tseng et al., 2019). Digital devices now enable continuous interaction and access to services anytime and anywhere (Abeele et al., 2018). However, constant connectivity often generates discomfort or a sense of loss when disconnected—a phenomenon referred to as technological unconsciousness (Thrift, 2004). Hoehe and Thibaut (2020) emphasized that individuals must be intentional, creative, and mindful in their technology use to enhance relationships and productivity; otherwise, technology risks eroding rather than supporting well-being.

Today, pervasive internet access has become so embedded in everyday life that most individuals perceive it as normal and harmless. Yet such normalization cannot be justified without acknowledging the risks associated with excessive technology use. While digital tools provide convenience and efficiency, they also introduce challenges that influence both individual and collective well-being (Dienlin and Johannes, 2020; Orben and Przybylski, 2019). Undoubtedly, technology has indeed improved many aspects of life, but its misuse can generate harmful behavioral patterns that deserve equal attention.

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Technology was originally developed to enhance the quality of human life (Connolly et al., 2016) and serves as a key determinant of overall life satisfaction (Kryzhanovskij et al., 2021). As Hoehe and Thibaut (2020) noted, technology itself is value-neutral—it can be beneficial or harmful depending on how it is applied. When technology intended to improve life is used without restraint, it can ultimately reduce quality of life. Since quality of life is closely tied to well-being (Upton and Upton, 2015), scholars have begun to recognize the need for a new construct —digital well-being—to describe well-being in the digital age. Yet, how individual conceptualize and experience digital well-being (DWB) remains insufficiently understood (Nguyen et al., 2022).

The emergence of the term "*digital well-being*" reflects a broader historical evolution in how societies interpret the relationship between technology and quality of life. Early scholarship on internet adoption (Kraut et al., 1998) and technostress (Tarafdar et al., 2007; Weil and Rosen, 1997) laid the conceptual groundwork by examining the psychological costs of rapid digitization. These studies highlighted the paradox of technological progress—its simultaneous capacity to enhance connectivity and generate new forms of strain.

Building upon this foundation, the notion of DWB emerged in the late 2010s as scholars and industry leaders sought to redefine a healthy relationship with technology.

Major corporate initiatives further popularized the concept. Google's Digital Well-Being and Apple's Screen Time programs, for instance, framed digital balance as a design priority, emphasizing user awareness and self-regulation. These industry efforts marked a pivotal shift from merely preventing overuse to actively designing for well-being, signaling a convergence between technological ethics, public health, and user-centered innovation. Thus, DWB represents both an academic and applied response to the evolving socio-technological landscape—bridging earlier studies on technostress and contemporary frameworks that emphasize agency, mindfulness, and design responsibility.

Ghai et al. (2023) examined global research trends using bibliometric approach, revealing that DWB began to emerge as a major topic in 2016 and has grown significantly since 2019. This growth correlates with global digital adoption during and after the pandemic, advances in technology, increased affordability of devices, and expanding digital infrastructure enabling broader connectivity. Ghai et al. (2023) continue to categorized DWB research into three domains: (1) Studies exploring its relationship with smartphone, social network, wellness, overuse, disconnection, and digital ethics, (2) Studies investigating how digital devices use related to individual well-being, and (3) Research focusing on social media's role in shaping DWB.

The overlapping use of related terms—such as digital wellness, digital mindfulness, digital minimalism (Dadischeck, 2021), digital health (Smits et al., 2022), digital ethics (Burr et al., 2020), and digital disconnection (Abeele et al., 2022; Jorge et al., 2022; Nguyen et al., 2022) —has blurred conceptual boundary. To advance the discourse, scholars must clearly distinguish DWB from adjacent constructs. Although several definitions exist, consensus on its fundamental aspects remains elusive. Öste (2020) viewed DWB as recognizing the digital self—a conscious relationship with technology. Abeele (2021a) described it as the optimal balance between being connected and disconnected, influenced by personal, device-related, and situational factors. Gui et al. (2017a) expanded the concept to include community dimensions, suggesting that collective norms, values, and expectations also determine DWB. Despite those contributions, a comprehensive and widely accepted definition remains lacking—particularly one that explicitly links DWB to conventional well-being concept.

Major technology companies have also adapted the term. For example, Google defines digital well-being as a pleasant state that arises when people use technology to achieve their goals. Through its dedicated website (https://wellbeing.google/), Google promotes awareness of how technology should empower, rather than distract, users from meaningful goals. Similarly, most major technology firms now allow users to personalize digital environment (Hutmacher and Appel, 2022). Features such as algorithmic filtering, personalized recommendation, and adaptive interface are designed to enhance user satisfaction. However, such personalization typically targets hedonic outcomes—maximizing pleasure and minimizing discomfort—rather than eudaimonic outcomes, which involve meaning and self-realization. True digital well-being thus engagement with purposeful, growth-oriented experiences (Hutmacher and Appel, 2022).

Another source of confusion arises from measuring of DWB through built-in applications that monitor screen time or limit usage (García-Hermoso et al., 2020; Przybylski and Weinstein, 2017; Twenge, 2019; Whitlock and Masur, 2019). These tools lack standardized criteria for determining optimal digital activity levels, as the impact of screen time varies across individuals (Przybylski and Weinstein, 2017). Moreover, empirical studies reveal inconsistent effects of such tools on well-being (Beattie and Daubs, 2020; Griffioen et al., 2021; Roffarello and De Russis, 2019). Ironically, users often rely on their devices to help them disconnect—adjusting app setting or selectively disabling restrictions (Harmon and Duffy, 2023). Additionally, the use of the term *well-being* in these applications is conceptually vague, as many exclude psychological and emotional dimensions essential to assessing well-being (Brey, 2015; Smits et al., 2022). Consequently, these tools serve primarily as self-monitoring system whose long-term impacts remain uncertain.

Given the complex interplay between humans and technology, there is pressing need for empirical synthesis to clarify this relationship and prevent conceptual stagnation (Abeele, 2021a). To achieve this, the present study adopts a scoping review approach to systematically collect, analyze, and synthesize existing evidence on DWB—its definitions, measurements, and interventions. By mapping the current literature, this review aim to establish a comprehensive conceptual foundation that can guide future empirical investigations and theoretical development.

## 2. Material and methods

A comprehensive understanding of digital well-being (DWB) is urgently needed, yet synthesizing international findings remains challenging due to the absence of systematic efforts to consolidate and map the existing literature. To address this gap, we conducted a scoping review designed to explore the current body of knowledge, identify conceptual and methodological patterns, highlight research opportunities for future studies. The scoping review was chosen for its suitability in mapping broad and evolving fields, such as DWB, that encompass diverse research traditions and methodologies. This approach allows for the integration of heterogeneous evidence without the constraints of meta-analysis, thereby providing a panoramic view of definitions, measurements, and interventions related to DWB.

The review followed the Joanna Briggs Institute (JBI) methodological framework for scoping review (Peters et al., 2020a) and adhered to PRISMA-ScR (Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews) guidelines (Tricco et al., 2018). The JBI Framework was selected for its flexibility and comprehensiveness in mapping emerging constructs. It emphasizes (1) a clearly defined review question encompassing participants, concept, and context; (2) systematic and transparent search strategies; (3) structured data charting; and (4) synthesis of findings into conceptual of thematic maps. Unlike systematic reviews, scoping reviews do not require critical appraisal or meta-synthesis (Tricco et al., 2018) which aligns with the exploratory nature of this study. By applying the JBI process, this review aimed to ensure transparency, reproducibility, and rigorous documentation across all stages, from database searching to data extraction and synthesis.

### 2.1. Review questions

The overarching goal of this scoping review was to extract, organize, and synthesize the current state of knowledge regarding digital well-being (DWB) across populations and contexts. Specifically, this review sought to identify:

1. Definitions — How has DWB been conceptually defined in the literature?
2. Measurements — What measurements have been developed to assess DWB?
3. Interventions — What interventions strategies have been implemented to promote DWB?

Together, these guiding questions were intended to produce a comprehensive conceptual map that can serve as a foundation for subsequent empirical research and theoretical refinement of DWB.

### 2.2. Eligibility criteria

The study employed the Population, Concept, and Context (PCC) framework recommended by JBI to determine the inclusion and exclusion criteria (Peters et al., 2020a; Pollock et al., 2023). This framework facilitated structured decision-making when selecting relevant studies and ensured alignment between the research question and the review scope. Inclusion criteria encompassed: (1) Peer-reviewed publications and grey literature explicitly addressing definitions, measurements, or intervention strategies of digital well-being; (2) Studies employing quantitative, qualitative, mixed-method, or review methodologies; (3) Publication written in English and available in full-text form; and (4) Research published from inception up to February 2024. The exclusion criteria included: (1) Articles not grounded in the social sciences or humanities (e.g., purely technical or engineering studies); (2) Papers presenting non-empirical, opinion-based, or conceptual-only perspectives lacking analytical data; and (3) Conference abstracts, preprints, technical reports, and workshop presentations without full-text access. Table 1 summarized the inclusion and exclusion criteria used in this scoping review.

### 2.3. Search strategy

To ensure methodological transparency and reproducibility, a comprehensive search protocol was developed in accordance with JBI framework (Peters et al., 2020a; Pollock et al., 2023). The search was conducted between 1 February and 20 March 2024 across seven electronic databases: Scopus, SAGE, ScienceDirect, ProQuest, PubMed, JSTOR, and Google Scholars. Following JBI recommendations (Aromataris and Riitano, 2014), the process included: (1) developing an initial logic grid aligned with the review questions, (2) identifying key terms and synonyms for digital well-being, (3) utilizing database-specific subject headings and field codes, (4) applying Boolean operators (“AND”, “OR”) to refine the search, and (5) iteratively testing and refining search results for accuracy and relevance.

**Table 1**
Inclusion and exclusion criteria used in the scoping review.

| Criteria | Description |
|---|---|
| Inclusion | Articles addressing digital well-being definitions, measurements, or intervention strategies<br>Peer-reviewed or grey literature publication (through handsearching, library database and google scholars)<br>Empirical or review-based methods<br>English language<br>Full-text is free accessible<br>Publication up to February 2024 |
| Exclusion | Derives from non-social-humaniora standpoint<br>Non-empiric results, or opinion-only works<br>Conference abstracts, preprints technical reports, or presentation lacking full text |

The full search string combined Boolean operators and synonyms to capture the breadth of terminology: (“digital well-being” OR “digital wellbeing” OR "mobile well-being" OR "mobile wellbeing" OR “online well-being” OR “online wellbeing” OR “smartphone well-being” OR “smartphone wellbeing” OR “social media well-being” OR “social media wellbeing”).

This query retrieved articles containing the term *digital well-being* and its lexical variants within titles, abstracts, or indexed keywords. Equivalent search strings were adapted for other databases such as SAGE, ScienceDirect, PubMed, ProQuest, and JSTOR, with syntax adjusted according to each platform’s indexing rules. For example, the scopus query was: TITLE-ABS-KEY (("digital well-being") OR ("digital well-being") OR ("mobile well-being") OR ("mobile wellbeing") OR ("online well-being") OR ("online wellbeing") OR ("smartphone well-being") OR ("smartphone wellbeing") OR ("social media well-being") OR ("social media wellbeing")) AND (LIMIT-TO (LANGUAGE, "English")) AND (LIMIT-TO (SUBJAREA, "PSYC") OR LIMIT-TO (SUBJAREA, "SOCI") AND (EXCLUDE (DOCTYPE, "re") OR EXCLUDE (DOCTYPE, "ed") OR EXCLUDE (DOCTYPE, "ch")).

This search string was designed to retrieve peer-reviewed articles written in English that explicitly discuss digital well-being (DWB) and its conceptual variations (e.g., online, mobile, smartphone, or social media well-being). The query was limited to relevant subject areas—including psychology and social, to ensure conceptual focus. Non-empirical such as reviews, editorials, and book chapters, were excluded to enhance methodological consistency. Equivalent search strings were adapted for other databases with syntax modifications specific to each indexing system.

All retrieved records were exported to Rayyan, a web-based systematic review tool, where both automatic and manual duplicate removal was performed. Two reviewers independently screened titles and abstracts, followed by full-text assessments based on the inclusion and exclusion criteria summarized in Table 1. Discrepancies were resolved through discussion and consensus to maintain rigor and inter-rater reliability. Reference lists of included studies were manually screened to identify additional sources, including relevant grey literature. This multi-stage process ensured that the final dataset reflected a comprehensive and methodologically transparent synthesis of the current state of research on digital well-being. As illustrated in Fig. 1, A PRISMA-ScR flow diagram summarized 1.693 records identified, 416 duplicates were removed, 1.277 records were screened, and 26 articles met all inclusion criteria for final synthesis.

### 2.4. Data extraction and charting

Data extraction followed JBI’s iterative approach, allowing for flexibility as new insights emerged during analysis (Pollock et al., 2023). A structured data charting form was designed to ensure consistency in data extraction. Each study was coded for author(s), publication year, country of origin, sample, research design and methodology, conceptual or theoretical framework, target population or context, primary objectives and key findings, and evidence of definitions, measurements, or interventions of digital well-being.

Data extraction was conducted iteratively—allowing for refinement of categories as new insights emerged—to maintain alignment with the review objectives. This process produced an evidence base for constructing an analytical summary and identifying research gaps. Two reviewers cross-checked the charted data to maintain consistency and inter-rater reliability.

### 2.5. Data analysis and synthesis

Consistent with the JBI methodology, this study did not perform a formal critical appraisal or risk-of-bias assessment, as the primary aim

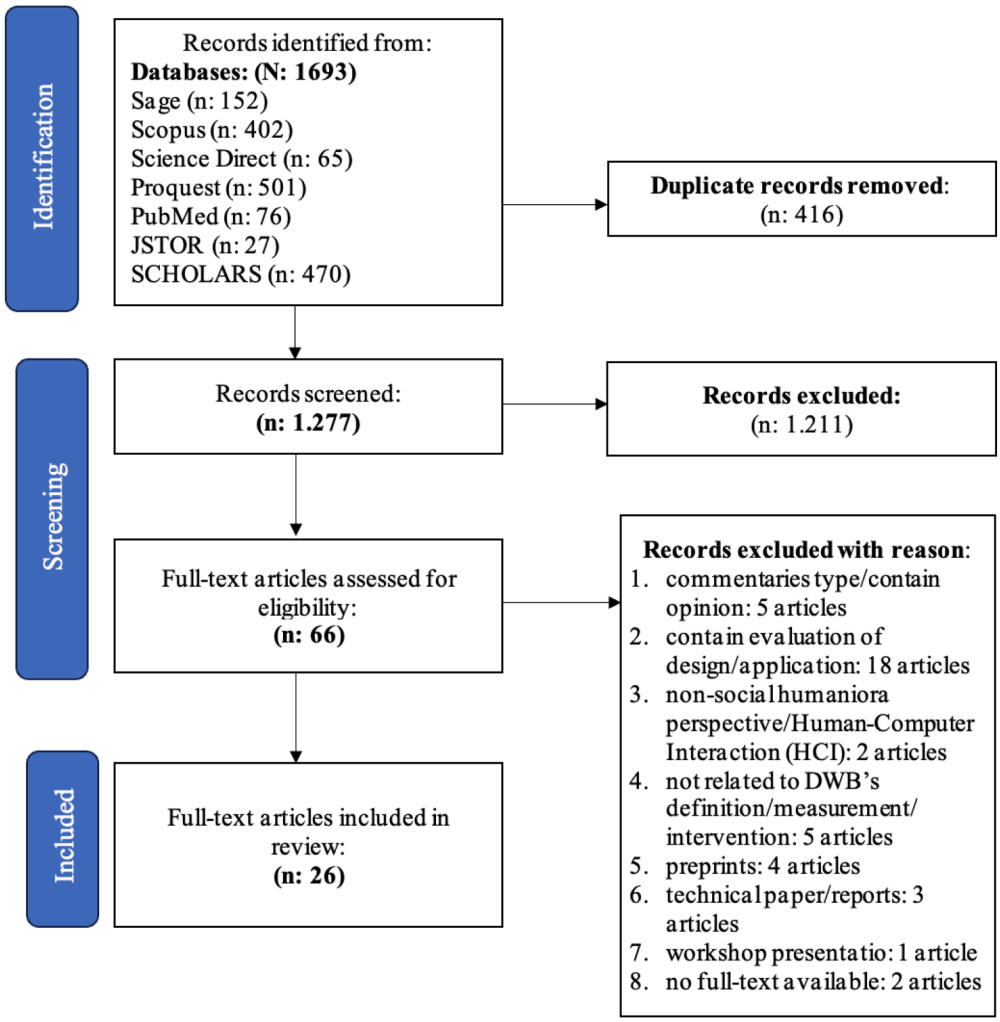


**Fig. 1.** PRISMA flow of study selection for digital well-being scoping review.

was to map the breadth and nature of evidence rather than evaluate effectiveness. However, several measures were implemented to enhance the reliability and credibility of synthesis result, such as: (1) dual independent screening minimized subjective bias; (2) transparent documentation of the inclusion process ensured traceability; (3) cross-verification of extracted data among authors maintained analytical consistency; and (4) reflexive discussion sessions were held the trustworthiness and transparency of the review while remaining consistent with the exploratory intent of scoping review.

The data were analyzed using thematic synthesis (Thomas and Harden (2008), involving three stages: (1) line-by-line coding of relevant text segments to identify recurring concepts; (2) development of descriptive themes summarizing these patterns; and (3) generation of analytical themes that captured the underlying conceptual relationship across studies. This dual-level synthesis enabled both descriptive mapping and interpretive insight. Findings are presented as a narrative synthesis supported by comparative tables and thematic mapping, structured around the three focal areas—definitions, measurements, and interventions.

### 2.6. Reproducibility

All methodological steps, including full search queries, timeframes, inclusion/exclusion criteria, and data charting templates, have been retained for potential replication and are available from the corresponding author upon reasonable request. This ensures that future scholars can reproduce or extend the current review process.

### 2.7. Ethical considerations

This scoping review involved analysis of publicly available data and did not require direct interaction with human participants. Nonetheless, ethical clearance was obtained from the Ethics Commission of the Faculty of Psychology, Universitas Gadjah Mada (Approval No. 1284/UN1/FPSi.1.3/SD/PT.01.04/2024) to maintain institutional compliance and transparency in data handling and reporting.

## 3. Results

Research on digital well-being (DWB) has expanded rapidly since 2016 (Ghai et al., 2023), reflecting the broader digital transformation influencing almost every aspect of human life. Key drivers include global internet penetration, the COVID-19 pandemic, widespread availability of affordable digital devices, and advances in digital infrastructure enabling faster connectivity.

From this study, a total of 26 studies were included, as summarized in the PRISMA-ScR flow diagram (Fig. 1). These studies represent diverse disciplines —including psychology, communication, public health, sociology, and information technology—illustrating the multidimensional nature of DWB research. The included publications represent diverse methodological, consist of: 7 qualitative, 5 quantitative, 4 mixed-method, and 10 conceptual or review-based. This distribution

reflects the methodological pluralism in DWB research where theoretical, empirical, and applied approaches frequently overlap.

The studies originated primarily from Europe ($n$ = 19), North America ($n$ = 1), Africa ($n$ = 1; Zimbabwe), Asia ($n$ = 4; Singapore, Thailand, China, Qatar), and Oceania ($n$ = 1; Auckland). While this pattern indicates strong scholarly engagement in Western contexts, the apparent geographic imbalance should be interpreted with caution. The review included only English-language, peer-reviewed sources indexed in major databases; therefore, the limited representation of Asian research may reflect database coverage and language bias rather than a genuine lack of scholarly activity in these regions. Moreover, this imbalance suggests that perspective from Asian population, despite their high internet penetration, remain underrepresented.

A detail summary of methodological characteristics —including country, theoretical orientation, participant group, and study design—is presented in Table 2. Overall, the body of evidence includes a balanced mix of qualitative, quantitative, mixed-method, and review-based studies, reflecting the field's early exploratory stage.

## 4. Discussion

To ensure the transparency, this section separates descriptive synthesis (reporting what studies found) and interpretive analysis (author's conceptual synthesis).

**Descriptive synthesis**

To ensure transparency in reporting, this section first presents a descriptive synthesis of the reviewed studies, summarizing their empirical characteristics, definitions, and methodological orientations without interpretation. This descriptive focuses synthesis of what studies reported, such as: conceptualizations of digital well-being (DWB), measurement instruments, and intervention, serving as the evidential foundation for the subsequent interpretive analysis.

### 4.1. Definitions of digital well-being

The need for the importance of technology is undeniable. In digital era, various kind of technology forming as digital media, tools or application is crucial in people's everyday life. The convenience provided has become an important force in everyday life. Yue et al. (2021) express that digital well-being (DWB) as a comprehensive word that includes multiple aspects of digital life.

As we begin to discovering DWB, we focus on how scholars had defined it. In this study, a variety of definitions were identified; however, certain articles share a commonality in that they associate DWB: (1) as the emerging concept of well-being; (2) as the impact of technology on well-being; and (3) as balanced-use strategy in a digital environment.

#### 4.1.1. Digital well-being as an emerging concept of well-being

Although well-being has long been a focus of psychological and social inquiry, its conceptualization within the digital era is still developing. Several scholars interpret DWB as an extension of conventional subjective well-being within technology-mediated environment. Arslankara et al. (2022) and Büchi (2021) define DWB as individuals'

**Table 2**
Characteristics of the 26 studies included in the review.

| Author | Year | Country | Background Theory | Inclusion Criteria Meet | Design Methodology | Respondents |
|---|---|---|---|---|---|---|
| (Abeele and Nguyen, 2022) | 2022 | Belgium | Communication | Definition | Literature Review | N/A |
| (Abeele and Nguyen, 2023) | 2023 | Belgium | Communication | Definition & Measurement | Cross-sectional survey | 1000 belgian adults |
| (Abeele, 2021a) | 2021 | Belgium | Communication | Definition & Intervention | Literature Review | N/A |
| (Al-Mansoori et al., 2023) | 2023 | Qatar | Human Behavior and Emerging Technologies | Definition | Scoping Review | N/A |
| (Gennari et al., 2023) | 2023 | Italy | Human-Computer Behavior | Definition | Qualitative | 24 high-school and teachers |
| (Arslankara et al., 2022) | 2022 | Turki | Education and Lifelong Learning | Measurement | Quantitative | 367 respondents |
| (Büchi, 2021) | 2021 | Switzerland | Social | Definition & Framework | Qualitative-Theory Building | N/A |
| (Burr et al., 2020) | 2020 | Oxford | Social Engineering | Definition | Thematic Review | N/A |
| (Cao and Li, 2023) | 2023 | China | Public Health | Definition Measurement Intervention | Scoping Review | N/A |
| (Dennis, 2021) | 2021 | Netherlands | Ethics and Information Technology | Intervention | Literature Review | N/A |
| (Gui et al., 2017b) | 2017 | Italy | Sociology of Education | Definition & Framework | Literature Review | N/A |
| (Kirby et al., 2017) | 2017 | England | Contemporary Ergonomics and Human Factors | Definition | Quantitative | 99 participants |
| (Gomes et al., 2023) | 2023 | Italy | Theories and Research in Education | Measurement | Quantitative | Age groups of 21 to 30 years old |
| (Mutsvairo et al., 2023) | 2023 | Zimbabwe | Media Communication | Definition | Qualitative | 10 heavy users |
| (Nguyen, 2021) | 2021 | Zurich | Mass Communication and Society | Intervention | Qualitative | 30 adults |
| (Nguyen et al., 2022) | 2022 | Switzerland | New media & society | Intervention | Quantitative | 1163 Swiss Internet users |
| (Ong et al., 2021) | 2021 | United Kingdom | Psychology | Measurement | Scoping Review | N/A |
| (Priyanka, 2023) | 2023 | Auckland | Social/Technology | Measurement | Mixed-method design | 20 informants and 280 respondents |
| (Roffarello and De Russis, 2023) | 2023 | Italy | Computer Science | Intervention | Case study | 93 case study |
| (Rosič et al., 2024) | 2024 | Belgium | Children and media | Measurement | literature review | 161 & 1171 Slovenian adolescents |
| (Saindon, 2021) | 2021 | Kentucky | Arts & Science | Definition | Mixed-Metho | 18 participants |
| (Schneider et al., 2022) | 2022 | Germany | Mobile Media & Communication | Intervention | Multi-Method | N/A |
| (Stankov and Gretzel, 2021) | 2021 | Serbia | Information Technology | Definition & Intervention | Qualitative | N/A |
| (Stone, 2023) | 2023 | Dublin | Education | Definition & Intervention | Mixed Method | N/A |
| (Thianthai and Tamdee, 2024) | 2024 | Bangkok | Health | Definition | Qualitative | 30 respondent age 18–32 years old |
| (Yue et al., 2021) | 2021 | Singapore | Media and Culture | Definition & Framework | Literature Review | N/A |

subjective well-being within a digital context shaped by extensive media and technology use. Gui et al. (2017) describe it as maintaining subjective well-being amid pervasive digital communication. Al-Mansoori et al. (2023) and Kirby et al. (2017) emphasize its emergence alongside online interaction, while Stankov and Gretzel (2021) extend it to multiple domains—physical, emotional, intellectual, and vocational.

Collectively, these definitions suggest that digital well-being reflects the evolution of subjective well-being in a digitally mediated world, influences by individual's ability to sustain balance and satisfaction across multiple life domains.

#### 4.1.2. Digital well-being as the impact of technology on well-being

A second cluster of studies conceptualizes DWB in terms of technology's influence—both positive and negative—on quality of life. Gui et al. (2017b) argue that DWB support comfort, safety, and fulfillment at both individual and community levels, whereas technostress (Stankov and Gretzel, 2021) and continues connectivity (Abeele and Nguyen, 2022) can diminish well-being. Burr et al. (2020) frame DWB as an ethical question about "what it means to live a good digital life", while Saindon (2021) links it to holistic user wellness through digital devices.

In the workplace, DWB refers to how technology positively or negatively affects employees' well-being (Stone, 2023). Meanwhile, Thianthai and Tamdee (2024) and Arslankara et al. (2022) stress that digital practices contributing to physical, mental, and social well-being are central to DWB. Scholars agree that both positive and negative impacts are inseparable, necessitating an understanding of both sides.

Gennari et al. (2023) broaden the scope by noting that technology's pervasiveness affects not only on individuals, but also their social ecosystem, implying DWB is both a personal and societal construct. Accordingly, Gui et al. (2017b) describe DWB as a condition reached by individuals through their skills and by a community whose standards, beliefs, and expectations enable fulfilment and protect its members.

Despite varied definitions of DWB, these studies converge on the view that DWB capture the balance between the benefits and burdens of technology use across physical, mental, and social domains. Although digital technologies are designed to enhance well-being, they also present challenges.

#### 4.1.3. Digital well-being as balanced-use strategy in digital environment

A third interpretation portrays DWB as the ability to achieve balance in technology use. Al-Mansoori et al. (2023) define it as technology that fosters positivity, personal growth, and healthy engagement. Yue et al. (2021) highlight three focal areas; cultivating responsible digital relationship, recognizing both positive and negative effects of digital engagement, and learning self-regulation strategy. While Cao and Li (2023) and Abeele (2021a) attempt to synthesize the term "balanced" to indicate a balance between the advantages and negative aspects of digital usage. Mutsvairo et al. (2023) agreed that DWB is attained when someone has the capability to address the problems posed by digital technology to well-being and recognizes its benefits. An individual's capability is influenced by their perspective on what constitutes a balanced digital media use (Nguyen et al., 2022).

Mutsvairo et al. (2023) caution against equating DWB screen-time reduction alone; instead, they emphasize individual's strategic capabilities to manage technology use effectively. Scholars such as Gui et al. (2017b) and Stone (2023) describe these competencies as digital skills that help individuals regulate attention, mitigate information overload, and sustain focus. Abeele (2021a) adds that such capabilities depend on a constellation of personal, device, and contextual factors, reinforcing that DWB is both individually enacted and social shaped.

### 4.2. Digital well-being framework

Mutsvairo et al. (2023) highlight the mechanisms of DWB in a nation experiencing an increase in mobile users, especially among young adults, remain largely unexplored. We found various concepts related to the DWB framework and identified several scholars perspectives.

In simple term, Gomes et al. (2023) encompasses DWB into three dimensions, mental, emotional and physical. While Abeele and Nguyen (2023) classified digital media users into four subgroups—skeptical, ambivalent, enthusiastic, and indifferent—based on their experiences of DWB, including their sense of control and perceived functional support. Focusing on young adults, Mutsvairo et al. (2023) point out four main insight regarding DWB, such as: (1) mobile phone use is deeply integrated into everyday life; (2) generational conflicts arise, as older generation encourages less device use; (3) there is ambiguity in distinguishing between normal usage and addition; and (4) digital media are seen both as tools of freedom and as sources of digital coercion with associated risks.

The first framework we found is conceptualized by Büchi (2021). A general framework of DWB focuses on the relationship between digital practices, harms/benefits, and well-being. It emphasizes that digital media itself is neither inherently harmful nor beneficial, but its impact depends on societal digitization. It highlighted three key interdependencies including: (1) social and technological opportunities and limits affect digital practices; (2) digital practices can cause both harms and benefits, and these often happen together; and (3) the overall balance of harms and benefits determines well-being (Büchi, 2021). Inline with this, Mutsvairo et al. (2023) claimed that digital media use is part of DWB, taking into account both subjective perception (micro--dynamics) and social and cultural environment (macro-dimensions). This concept indicates that DWB pertains to both individual and societal dimensions.

In another perspective, scholars discuss issues within the DWB paradigm. Burr et al. (2020) linked DWB into three broader theme such as positive computing, personalised human-computer interaction and autonomy with self-determination. More precisely, Yue et al. (2021) highlights nine key dimensions of DWB namely digital safety and security, digital rights and responsibilities, digital health and self-care, digital creativity, digital emotional intelligence, digital communication, digital consumerism, digital employment and entrepreneurship, and digital activism/civic engagement. Thus, three additional dimensions are introduced in the framework to reflect this relationship: digital skills, identity, and empowerment and agency. Meanwhile, Thianthai and Tamdee (2024) points out four digital themes namely: maintaining a balanced use of technology (digital usage), building social awareness through digital literacy; (3) managing negative emotions through social relationship in digital networks; and (4) continuously learning and education oneself in the digital era (digital learning).

Building on the previous explanation, we acquire insights into the DWB framework; however, a structured approach or conceptual model that delineates the key dimensions, relationships, and interactions inherent in DWB requires further development.

### 4.3. Measurement of digital well-being

Measurements of DWB in the digital context emphasise well-being itself, while the digital aspect remains unquantified (Cao and Li, 2023). Rosič et al. (2024) and Ong et al. (2021) claimed that instruments for measuring DWB are scarce. Nonetheless, five instruments were identified. First, Abeele and Nguyen (2023) developed an 11-item self-constructed scale assesing perceived agency and functionality of digital media in relation to well-being, daily goals, and social life. Second, Arslankara et al. (2022) created a 12-item DWB scale with three factors—digital satisfaction, safe and responsible behavior, and digital wellness. Third, Gomes et al. (2023) designed a multidimensional DWB scale encompassing social, physical, mental, emotional dimensions, but finalized with only mental, emotional, physical dimensions due to the social dimension's correlation with mental and emotional health. The study relies on self-reported data, so response bias may be present. Fourth, Priyanka (2023) qualitatively identified DWB as a multi-state construct involving physical care, emotional resilience, agency, and

communion, suggesting that cognitive and affective appraisal are essential for achieving DWB. Fifth, Rosič et al. (2024) developed The Perceived Digital Well-Being in Adolescence Scale (PDWBA) targeting social, cognitive, and emotional domains for youth population.

Across instruments, DWB consistently appears as a multidimensional construct encompassing mental, emotional, physical, social, and behavioural aspects. However, no existing instruments yet captures these dimensions in an integrated or validates framework. Most rely on self-report, which introducess subjectivity and limit cross-context comparability. Ong et al. (2021) stress the need to distinguished online well-being from subjective well-being measures and to reach consensus about what constitutes the "online world". Consequently, future measurement development must address both conceptual precision and psychometric robustness.

*To enhance methodological clarity and offer practical value, we conducted a comparative analysis of all identified DWB instruments. This synthesis aims to illustrate how existing instruments operationalize the construct differently across studies. We summarized the* comparative measurement analysis in Table 3.

Although the five instruments collectively advance the operationalization of DWB, they reveal several conceptual gaps. Notably, none of the existing scales integrated socio-cultural determinants of eudaimonic dimensions that reflect personal growth, purpose, and meaning in digital engagement. Most instrument remain confined to self-reported behavioral indicators of hedonic evaluations (e.g. satisfaction, comfort, or perceived control), overlooking contextual and relational dynamics that shape digital experiences.

### 4.4. Intervention of digital well-being

Abeele (2021a) expressed that constellations (personal, device, and context-specific factors) create paths to DWB that change outcomes for well-being when they are repeated, and the effectiveness of interventions depends on how they mess up these pathways. We found intervention for DWB in a variety of settings, such as schools, workplaces, and tourism. However, most of the interventions we often encounter are related to digital disconnection (Abeele et al., 2022; Abeele and Nguyen, 2022, 2023; Nguyen et al., 2022). In the part that follows, we will attempt to provide a more thorough explanation of our findings.

#### 4.4.1. Digital disconnection

Disconnection strategies are a complex endeavour that involve cultivating self-awareness around social media consumption and undertaking "digital detoxes" by disengaging from devices, platforms, and apps (Nguyen, 2021). Abeele and Nguyen (2023) found that individuals employ various digital disconnection strategies, which can be categorized into individual, group, and institutional levels; however, their prevalence varies significantly. The impact of digital disconnection on well-being can not be generalized nor straightforward since making disconnection from these devices is a complex endeavour (Abeele and Nguyen, 2022). The choice and impact of disconnection strategies may be influenced by individual, technological, and situational factors, such as social presence and engagement in concentration-demanding activities (Abeele and Nguyen, 2023).

Disconnection strategies can facilitate control of information flow and communication via social media, influencing online experiences and the related advantages. Nguyen (2021) argue that disconnecting was favourable for psychological (peace and quietness), physical (better sleep), and productivity and time management reasons. Many of those who temporarily disengaged from digital media believed that by reconnect could improve their digital experiences and well-being. In contrary for some users, disconnecting did not help some individuals because they felt social media did not give them any damage. However, digital users generally were happy to spend less time on something that they felt no longer valuable.

Previous studies have shown that a persistent age difference influences the disconnection strategy (Nguyen et al., 2022). In terms of disconnection strategies, older generations were more likely to use rule-based strategies (such as taking technology breaks), but younger generations were more likely to rely on device and app features (such as turning off notifications). Younger generations are more likely to disconnect because of concerns about digital content and privacy, signalling that concerns about personal data may become less important as a reason to disconnect in the future. Furthermore, younger generations are more likely to use app features to disconnect than older generations.

Nguyen (2021) expand the theory of disconnective behaviour consist of three main concern, such as: (1) disconnecting from social media is typically temporary and context-dependent, occurring across various levels, such as detaching from devices, social media platforms, or specific features, content, or individuals; (2) concepts from digital inequality research provide insight into how people adopt disconnection strategies; and (3) disconnection methods enable individuals to regulate the flow of information and communication on social media, influencing their online experiences and the advantages they gain from it.

#### 4.4.2. Rule and characteristic-based strategy

Nguyen et al. (2022) stated that balanced digital media use is primarily concerned with subjectively appropriate amounts of use, purposeful use, social connections, non-addiction, and time for “real life.

**Table 3**
Comparative analysis of five digital well-being measurement instruments.

| Instrument/Author (Year) | Core Dimensions | Population / Sample | Psychometric Properties | Distinctive Contribution | Limitations |
|---|---|---|---|---|---|
| Digital Well-Being Scale (Arslankara et al., 2022) | 3 dimensions: digital satisfaction, safe and responsible behavior, and digital wellness. | Teachers and adult learners (Turkey, $n = 367$) | Cronbach's $\alpha = 0.88$; CFA confirms 3-factor model | Introduces behavioral and attitudinal components emphasizing digital responsibility | Limited cross-cultural validation; focuses primarily on educators |
| Digital Media Agency & Functionality Scale (Abeele and Nguyen, 2023) | 2 dimensions: perceived agency and functional support of digital media | Adults (Belgium, $n = 1000$) | Self-constructed; internal reliability $\alpha > 0.80$ | Highlights perceived *control* and *goal alignment* of digital use | Lacks multi-dimensional assessment; limited validation |
| Digital Well-Being Scale (Gomes et al., 2023) | 3 final dimensions: mental, emotional, and physical (originally 4, social merged) | Young adults (India, $n = 300$) | EFA/CFA support construct validity; $\alpha = 0.82$–$0.90$ | Provides multidimensional coverage with affective and physiological indicators | Social dimension underdeveloped; self-report bias |
| Perceived Digital Well-Being in Adolescence Scale (Rosič et al., 2024) | 3 dimensions: social, cognitive, and emotional | Adolescents (Slovenia, $n = 1171$) | Good reliability ($\alpha = 0.78$–$0.89$); correlates with self-esteem & self-control | Captures adolescent-specific experience and developmental context | Age-restricted; not tested in adult populations |
| Qualitative Digital Well-Being Model (Priyanka, 2023) | 4 dimensions: physical care, emotional resilience, agency, and communion | Mixed sample (New Zealand) | Conceptually derived; qualitative thematic validation | Integrates eudaimonic and relational dimensions of digital life | Lacks quantitative psychometrics; exploratory |

Type of motivations people have for disconnecting from digital media (i.e., well-being and availability-related motivations, content- and privacy-related motivations, and social motivations) and strategies people use for disconnecting (i.e., rule-based strategies and feature-based strategies). For the same reason we already said, older generations like rule-based strategies (like technology-free moments), while younger generations like features that are built into devices and apps (like turning off alerts). As we learned from earlier studies, we found at least two strategies connected with the cultivation of digital well-being: rule-based and characteristic-based.

Amid the COVID-19 pandemic, the utilization of technology and digital media surged significantly. A popular rule-based strategy entitled Digital Well-Being Guideline for Parents was published by The Center for Humane Technology (CHT). This guideline consist of set rules of screen time and online content in order to encouraging mindful usage, design tech-free zones and times, modeling healthy digital habit, utilizing parental control and monitoring tool, and promoting physical activity and offline engagement (Center for Humane Technology, 2020). Dennis (2021) explained more that the DWB guideline grouped into three categories namely abstinence, scepticism and mindfulness. However, Dennis (2021) argue that the problem of rule-based strategy—an extrinsic motivation strategy—countering persuasive technology radically underestimates its power. This places a significant deal of personal accountability for the user's digital well-being on them. Conversely, characteristic-based strategy seek to influence users' intrinsic motivation, enabling them to develop character qualities that promote their digital well-being. Furthermore, Dennis (2021) explained the practical extension of character-based strategy can be found in self-care app technology.

Both strategies offer an individualised approach to DWB which provides the condition to blame when we “break the rules”. Moreover, both strategies fail to recognize how online settings are intentionally designed to encourage behaviors that are detrimental to DWB, especially at a time when we are required to spend much of our time online. Therefore, both strategies simply not powerful enough to cultivate DWB in the face of the persuasive technologies that dominate today’s online environments. There was an argument called "situationism" that started this worry. In that debate, people thought that personal willpower and character traits were more important than environmental cues. Dennis (2021) comes with the idea to seek a third strategy. It expected to complement personal responsibility with an approach that addresses the environmental cues that guide individual behavior.

#### 4.4.3. Salutogenic strategy

An intervention was identified referred to as the Integrative of Mobile Media Use and Need Experience (IM3UNE). Rather than presuming mobile media to be harmful to health, the IM$^3$UNE adopts a salutogenic perspective. It emphasises strategies for maintaining health while engaging with mobile media extensively in everyday life. This model of strategy conceptualized by (Schneider et al., 2022) posits that both the satisfaction and frustration of basic psychological needs play a central role in how demanding mobile media use influences well-being. However, the effects of these needs depend on how individuals perceive, interpret, and respond to mobile media demands, which is shaped by their sense of coherence—a global life orientation. The research integrates prior findings and suggests that mindfulness, self-control, and a sense of meaning in life are key personal resources that strengthen one’s sense of coherence, helping individuals better manage mobile media demands.

By distinguishing between experiences of need satisfaction and frustration, the model offers a nuanced view of mobile media’s effects on well-being. Demands from mobile media can lead to either positive or negative outcomes, depending on how they are perceived. Specifically, individuals with high mindfulness, self-control, and meaning in life (i.e., a strong sense of coherence) are more likely to see mobile media demands as satisfying rather than frustrating, ultimately promoting well-being. Fig. 2 illustrates how this model functions.

#### 4.4.4. Various setting of intervention

*4.4.4.1. Education institution.* In education institution, understanding DWB as component of digital ability has a significant implication. Gui et al. (2017b) expand that the ability to manage communication overabundance should be integrated into learning system. Traditionally, the development digital skill limited to technically operate technologies or to teach using digital media. However, it now essential for the learning institution to focus on cultivating skills that limit and direct this usage. Schools play a significant role in fostering critical and balanced media use in daily life, extending beyond school hours. This type of intervention will increasingly a central role in addressing social inequality.

Educational solutions can enhance students' DWB and motivate them to develop technologies for social good. One of the interventions in the educational setting is *Benessere Digitale,’*i.e.,the university course through which we explored the potential impacts of education on students’ DWB and students’ attitudes towards tackling DWB problems and designing related solutions (Roffarello and De Russis, 2023). This cource aims to provide an understanding of DWB from both technological and psychological perspectives, teach methods to analyze and address DWB issues, enable students to design solutions that reduce negative impacts of excessive digital device use, and apply learned concepts through collaborative engineering projects. Furthermore, empirical study confirm that the course enhance’s critical thinking and perception of DWB, proposing innovative technological solutions beyond existing digital self-control tools.

Young children takes part in educational settings. Cao and Li (2023) found that the existing evidence highlights two effective strategies for enhancing young children’s DWB: digital applications and early interventions. Some studies emphasize early interventions designed to reduce screen time and promote well-being.

*4.4.4.2. Workplace.* Digital technology has both positive impacts and challenges. The positive impact consists of flexibility, autonomy, media multitasking, access to resources, and effective communication. In contrast, digital technology is causing challenges in work-home boundary management, digital overload, and digital distractions (Stone, 2023). Stone emphasized the dual-aspect concept of DWB intervention in the workplace, aiming to help workers address challenges and understand the potential positive effects of digital technologies. Digital skill training, micro-boundary, and disconnection strategy make up the approach. One of DWB intervention in workplace is “the deep work” strategy involves allocating specific time during the workday to disconnect from technology and focus on a single task.

*4.4.4.3. Tourism.* Stankov and Filimonau (2019) point out that tourism business should give more attention to their guests’ DWB by co-creating experiences that let them use technology in a balanced and thoughful way, which will encourage them to disconnect and interact with their surrounding more fully. This strategy of intervention giving a more enjoyable and memorable holiday moment. Mindfulness interventions in tourism must align with the primary motivations of guests in their pursuit of DWB. This can be based on hedonic motives, which emphasize happiness and pleasure, and eudaimonic motives, which prioritize meaning and self-realization. A few example of implementation such as: (1) giving gratification of mindfulness practices (if guest having amount of time spent not using smartphone) can be incentived via price discount, loyalty points or vouchers; (2) the silent or reflection corners equipped with tools that support “present moments”; or (3) the internet free zone. Guests may select this option if they believe it will be the most effective way for them to manage their smartphone usage time.

We agree that DWB in tourism is crucial to ensure the integration of digital media into the travel experience does not overwhelm tourists.

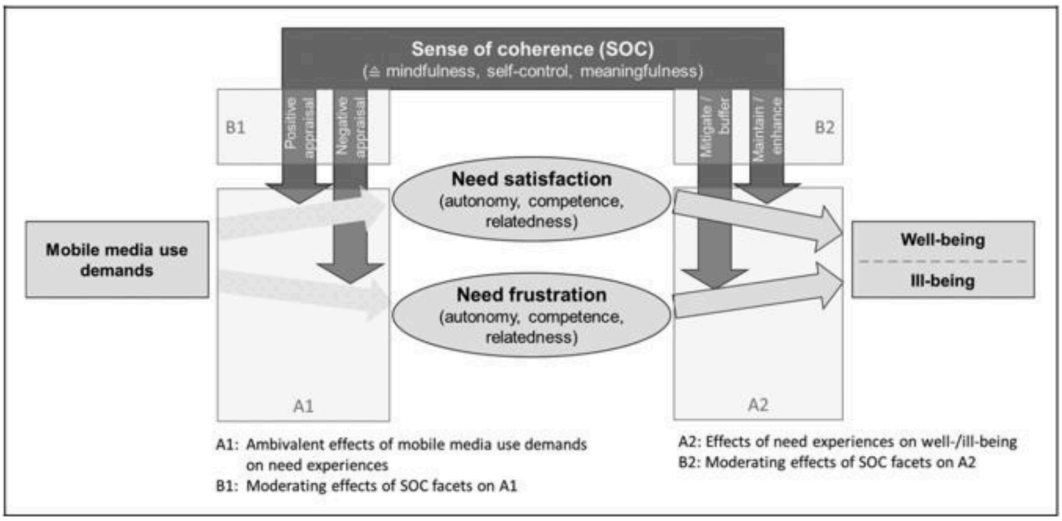


**Fig. 2.** Integrative model of mobile media use and need experience (IM³UNE) as proposed by Schneider et al. (2022).

Digital tools can improve the experience by giving useful knowledge, lowering stress, and encouraging mindfulness if they are used correctly. The tourism industry can help tourists stay healthy and happy while they're away by focussing on things like well-being, privacy, inclusion, and finding a balance between digital and physical experiences. When used in hospitality, mindfulness-based well-being interventions should take into account people's views, emotional states, and way of life, and they should be adaptable to meet the needs and expectations of different guests.

### 4.5. Interpretive synthesis

*Building upon the findings related to definitions, measurement, framework, and intervention presented earlier, we proceeded to integrate these insights to reveal broader thematic connections within the literature.* Through this interpretive synthesis, several recurring themes were identified and subsequently organized according to their conceptual relationships. Integrating all findings, three overarching themes were synthesized:

1. Behavior intention to use technology—emphasizes users' perceived ease, usefulness, and the ability to maintain optimal, balanced use. This reflects the self-regulatory and motivational aspects of digital behavior shaped by perceived control and purpose.
2. Eudaimonic well-being—represents the psychological fulfillment derived from technology use, encompassing self-actualization, self-satisfaction, and quality of life. It highlights how meaningful and mindful engagement can transform digital use into a source of personal growth.
3. State of socio-cultural—captures the influence of social norms, shared values, and role expectations on digital practices. It underscores that DWB is embedded within broader cultural and relational systems that shape individual experiences.

Together, these themes illustrate that DWB is a dynamic, multidimensional construct—neither purely psychological nor technological—but situated within a socio-digital ecosystem that integrates cognition, emotion, and culture. The interpretive thematic structure is illustrated in Fig. 3.

In summary, the inclusion of the comparative measurement analysis (Table 3) not only strengthens the methodological rigor of this review but also elevates it from a descriptive synthesis to a practical analytical resource. This comparative insight enables scholars to situate their research within existing paradigms, understand which instruments have been employed, recognize their limitations, and identify concrete pathways for methodological innovation.

Bridging these limitations, our interpretive synthesis emphasizes that a truly integrative framework of DWB must encompass three interrelated dimensions—(1) behavioral intention to use technology, (2) eudaimonic well-being, and (3) socio-cultural context. Together, these dimensions illuminate how individuals regulate their digital practices, derive meaning and fulfillment from technological engagement, and are shaped by broader cultural and social norms. Consequently, future measurement development should aim to embed these dimensions within a unified, multidimensional structure capable of capturing the full psychological and contextual complexity of human–technology interaction.

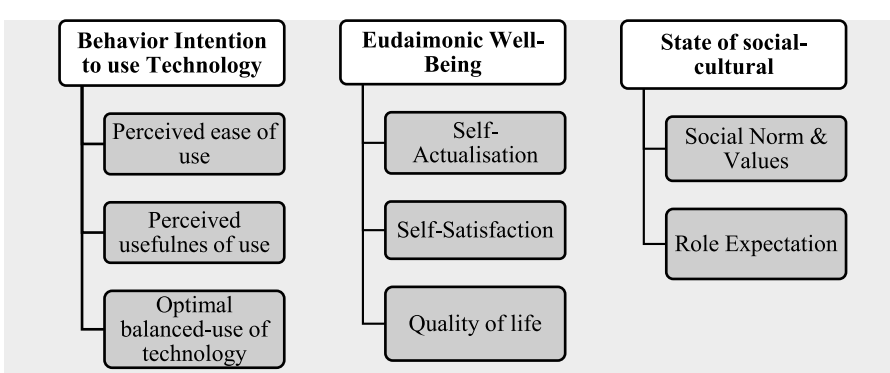


**Fig. 3.** Three overarching themes of digital well-being identified in the scoping review.

It should be acknowledged that the synthesis already contributes several valuable outcomes. This review proposes a novel three-themes conceptual framework linking behavioral intention, eudaimonic well-being, and socio-cultural context; identifies critical research gaps, including the underrepresentation of Asian scholarship; evaluates the inconsistent effectiveness of digital disconnection strategies; and highlights promising salutogenic approaches such as the IM³UNE model.

## 5. Conclusion

This scoping review synthesized current evidence on definitions, measurements, and interventions of digital well-being (DWB), revealing substantial progress yet persistent conceptual and empirical gaps. DWB represents an emergent, multidimensional construct that bridges human psychology and digital ecology. It encompasses not only hedonic aspects of comfort and satisfaction but also eudaimonic dimensions of meaning, balance, and growth within technology-mediated life.

Although the concept is increasingly recognized across disciplines, the field remains fragmented, with overlapping terminologies and inconsistent operationalization. Across 26 studies, three dominant conceptualizations emerged—DWB as the emerging concept of well-being, as the outcome of technology's impact, and as a balanced-use strategy to technology use. These perspectives collectively suggest that DWB is not merely a technological issue but a psychosocial construct rooted in both individual agency and socio-cultural context.

At a theoretical level, this review confirms that subjective and eudaimonic models of well-being remain highly relevant in digital contexts. The digital environment introduces new determinants—such as constant connectivity, algorithmic mediation, and platform design—that modify how individual pursue satisfaction, meaning, and self-regulation. As such, DWB can be viewed as an adaptive evolution of human well-being in response to pervasive technological mediation.

Despite conceptual advances, several critical gaps persists. First, the lack of consensus regarding DWB's defining attributes and dimensions hinders theoretical accumalation. Current framework (e.g. Buchi, 2021; Yue et al., 2021) offer valuable direction but remain partial and discipline-specific. Future work should aim to integrate psychological, sociological, and technological perspective into holistic, cross-disciplinary model. Second, the field still lacks standardized and psychometrically validated instruments. The five identified scales capture overlapping but distinct domains, reflecting the multidimensionality of DWB yet failing to converge on a unified construct. Establishing measurement equivalence across cultural contexts will be essential, especially given that most current tools were developed in Western samples. Comparative validation in Asian and Global-South populations could enrich the construct's universality and ecological validity. Third, findings on intervention effectiveness remain mixed. Most interventions emphasize digital disconnection or sefl-regulation, assuming that reduced exposure inherently improves well-being. However, such approaches risk oversimplifying the complex relationship between humans and technology. Evidence from salutogenic and educational interventions (e.g. IM³UNE model, Roffarello and De Russis, 2023) demonstrate that proactive engagement—cultivating mindfulness, purpose, and digital literacy—may yield more sustainable outcomes than abstinence-based strategies.

To stimulate future research and practice, scholars should pursue the next generation of DWB measures—tools that are comprehensive, context-sensitive, and culturally adaptive. Such instruments must capture both subjective experience and behavioral evidence, reflect regional and developmental diversity, and align with theoretical models that integrate individual, technological, and societal determinants. By advancing toward unified, empirically grounded measurement systems, researchers can more effectively map the dynamic interplay between human well-being and digital environments.

Finally, the geographic imbalance observed in this review reveals a critical representation gap. Although Asia hosts some of the world's largest online populations, research output from this region remains limited. This asymmetry constraints the cultural generalizability of existing theories and underscores the need for inclusive global collaboration in DWB research.

This review contributes to the emerging discourse by synthesizing DWB as a triadic construct encompassing: (1) Behavioral intention to use technology—users' perceived control, usefulness, and ease of achieving balance; (2) Eudaimonic well-being—personal growth, self-satisfaction, and purposeful engagement with technology; and (3) Socio-cultural context—collective norms, values, and expectations that frame digital practices. Together, these dimensions highlight DWB as both individual and systemic, shaped by interaction among personal competencies, technological design, and societal structure.

Practically, the findings offer implication for education, workplace, and policy. In educational institutions can embed DWB literacy and reflection within curricula to cultivate responsible and self-aware users. In workplace should promote micro-boundaries and mindful technology use to mitigate overload while sustaining productivity. For policy maker could develop regulatory framework encouraging technology companies to design well-being-centered digital environment, integrating persuasive design ethics and user-agency safeguards.

These insights align with the WHO's 2023 Global Digital Health Strategy, which emphasizes the inclusion of psychological well-being indicators in technology policy. Policymakers should therefore integrate DWB within national digital literacy and mental health frameworks, ensuring that human flourishing remains a central goal of digital transformation initiatives.

Given the global scale of digital technology use and its broad impact on mental health, social cohesion, and productivity, the findings of this review carry significant implications for multiple stakeholders:

1. Regulators and policymakers should establish frameworks that ensure technology serves public well-being. This includes mandating transparency in algorithmic systems, setting ethical design standards, and supporting research funding for digital health and well-being initiatives.
2. Platform designers and technology companies must embed human-centered design principles—including autonomy, inclusivity, and mindful engagement—into digital ecosystems to reduce harm and foster sustainable use.
3. Educators and academic institutions should integrate digital literacy and self-regulation strategies into curricula to equip individuals with the awareness and skills needed for balanced digital engagement.
4. Public health authorities should recognize DWB as a determinant of population health, incorporating it into prevention strategies, national surveys, and health promotion campaigns.

Advancing DWB requires shared responsibility and collaboration among researchers, policymakers, industries, and communities. Through collective commitment, digital systems can evolve from sources of distraction and stress into platforms that genuinely enhance human flourishing. Achieving DWB demands more than managing screen time—it requires reimagining how technologies align with human values. This review offers a foundation for theoretical, practical, and policy advancement, translating empirical insights into actionable guidance for building healthier, more equitable, and purpose-driven digital societies.

## 6. Future study of digital well-being

Despite its growing relevance, research on DWB remains unevenly distributed across regions and disciplines, with limited representation from non-Western contexts. To advance toward a coherent framework, future studies should:

1. Develop and validate comprehensive, culturally sensitive measurement instruments;
2. Pursue longitudinal and experimental designs to establish causal links between digital behaviors and well-being outcomes;
3. Explore strength-based and contextualized interventions beyond disconnection paradigms;
4. Investigate policy-level and ethical dimensions that shape digital environments and user agency.

Researchers should focus on intervention development to promote DWB. These interventions might include promoting self-actualization, balancing technology usage, and fostering supportive social norms in communities. The digital well-being apps, which are now the major tools that claim to maintain well-being in technology use, need to be re-evaluated. To understand the evolution of DWB over time, longitudinal studies are also necessary. Such research could provide valuable insights into how individuals' digital habits and socio-cultural norms change, influencing their overall well-being. Research on DWB policies at local, national, and international levels could guide the creation of healthier digital ecosystems, enhancing individuals' quality of life in a rapidly evolving digital era.

Ultimately, achieving DWB requires more than managing screen time—it demands a transformative understanding of how humans and technologies co-evolve. By embracing this integrative perspective, scholars and practitioners can design digital ecosystems that promote flourishing, balance, and meaningful connection in an increasingly connected world.

## Ethical statement

Hereby, I, Sovi Septania, assure that for the manuscript "Navigating Well-Being in The Digital Era: A Scoping Review of Digital Well-Being" is fulfilled:

- The work described has not been published previously except in the form of a preprint, an abstract, a published lecture or registered report. See our policy on multiple, redundant or concurrent publication.
- The article is not under consideration for publication elsewhere.
- The article's publication is approved by all authors and tacitly or explicitly by the responsible authorities where the work was carried out.
- If accepted, the article will not be published elsewhere in the same form, in English or in any other language, including electronically, without the written consent of the copyright-holder.

## Ethical clearance

Ethical approval for this scoping review has been granted by Ethical Commission of Psychology Faculty Gadjah Mada University with number 1284/UN1/FPSi.1.3/SD/PT.01.04/2024.

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## CRediT authorship contribution statement

**Sovi Septania:** Writing – original draft, Formal analysis, Data curation, Conceptualization. **Avin Fadilla Helmi:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Haidar Buldan Thontowi:** Writing – review & editing, Methodology, Conceptualization.

## Declaration of competing interest

The author declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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